The unit circle

1 State the quadrant in which the following angles are located:

a 299° **b** 5° **c** 160° **d** 229°

e 40° **f** 310° **g** -138° **h** -244°

2 For the given functions, state the following:

i The quadrants where the function is positive.

ii The quadrants where the function is negative.

a Sine function **b** Cosine function

c Tangent function

3 State whether the values of the following are positive or negative:

a $\sin 310^\circ$ **b** $\sin 50^\circ$ **c** $\sin 130^\circ$ **d** $\sin 230^\circ$

e $\sin 31^\circ$ **f** $\tan 31^\circ$ **g** $\cos 267^\circ$ **h** $\sin 267^\circ$

i $\cos 180^\circ$

4 Consider the given angles below and state whether the following are true or false:

$$w = 253^\circ, \quad x = 265^\circ, \quad y = 193^\circ, \quad z = -258^\circ, \quad a = -182^\circ, \quad b = -257^\circ$$

a Angles z, a, and b are in Quadrant 3.

b $\sin z$ and $\sin b$ are both negative.

c $\tan(z - 90^\circ)$ and $\tan(z + 90^\circ)$ are both positive.

d The angles w, x, and y have negative cosine values.

e Angles w, x, and y are in Quadrant 3.

5 State the quadrant where the point in each scenario is located:

a The point (x, y) is on the unit circle at a rotation of θ such that $\sin\theta = \dfrac{5}{13}$ and $\cos\theta = \dfrac{12}{13}$.

b The point $P(x, y)$ is on the unit circle at a rotation of θ such that $\sin\theta = -\dfrac{12}{37}$ and $\cos\theta = \dfrac{35}{37}$.

6 State the quadrant where the angle in eac[redacted]nario is located:

a θ is an angle such that $\sin\theta > 0$ and $\cos\theta < 0$.

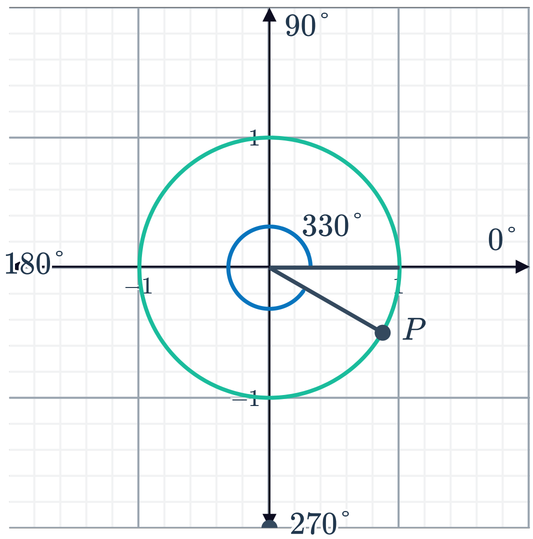
b θ is an angle such that $\tan\theta < 0$ and $\sin\theta > 0$.

c θ is an angle such that $\tan\theta < 0$ and $\cos\theta < 0$.

d θ is an angle such that $\tan\theta > 0$ and $\sin\theta > 0$.

Equivalent angles

7 Point P on the unit circle shows a rotation of 330°. Find the related acute angle in the first quadrant.



8 The diagram shows points $P(a, b), Q, R$ and S, which represent rotations of $49^\circ, 131^\circ, 229^\circ$ and 311° respectively around the unit circle.

a Find the coordinates of the following points in terms of a and b:

i Q ii R iii S

b Find the equivalent trigonometric ratio in the first quadrant of the following:

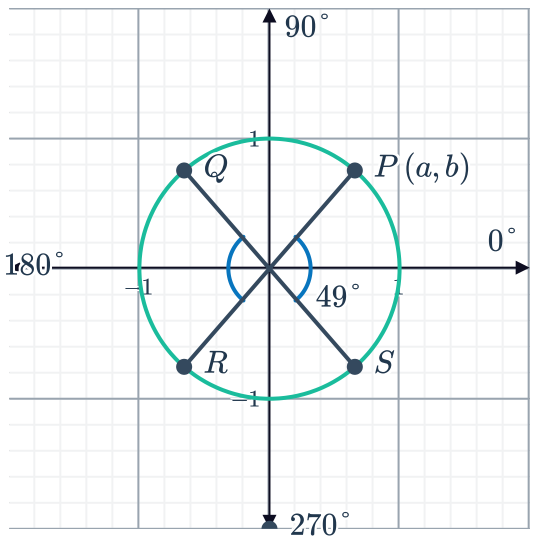
i $\sin 131^\circ$ ii $\sin 229^\circ$

iii $\sin 311^\circ$

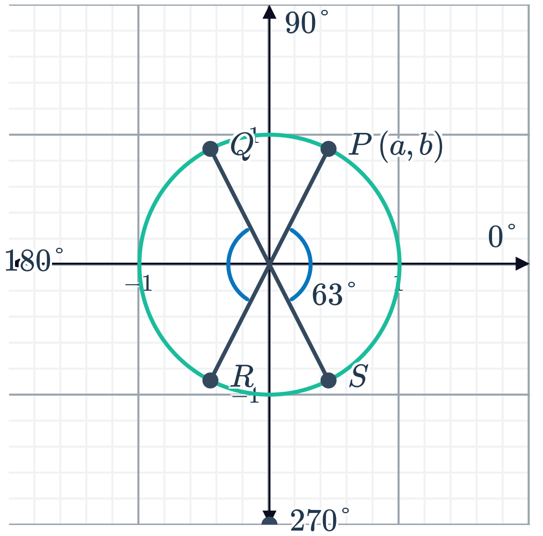
c Hence, express each of the following in terms of $\sin x$:

c $\sin(180^\circ - x)$ ii $\sin(180^\circ + x)$

iii $\sin(360^\circ - x)$



9 The diagram shows points $P\left(a,b\right), Q, R$ ar[redacted]which represent rotations of $63^\circ, 117^\circ, 243^\circ$ and 297° respectively around the unit circle.



a Find the coordinates of the following points in terms of a and b:

i Q ii R iii S

b Find the equivalent trigonometric ratio in the first quadrant of the following:

i $\cos 117^\circ$ ii $\cos 243^\circ$

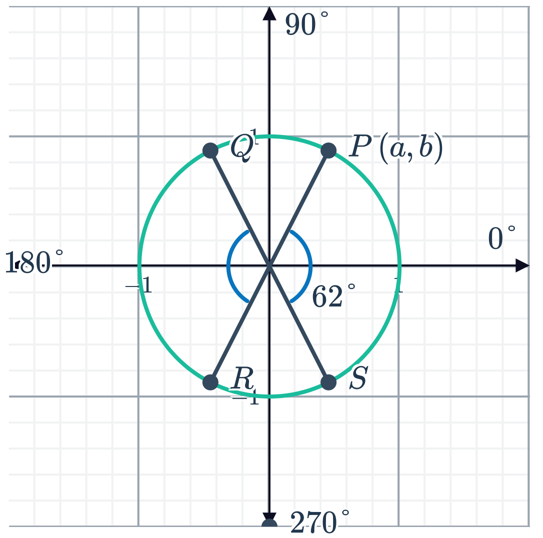
iii $\cos 297^\circ$

c Hence, express each of the following in terms of $\cos x$:

i $\cos\left(180^\circ - x\right)$ ii $\cos\left(180^\circ + x\right)$

iii $\cos\left(360^\circ - x\right)$

10 The diagram shows points $P\left(a,b\right), Q, R$ and S, which represent rotations of $62^\circ, 118^\circ, 242^\circ$ and 298° respectively around the unit circle.



a Find the coordinates of the following points in terms of a and b:

i Q ii R iii S

b Find the equivalent trigonometric ratio in the first quadrant of the following:

i $\tan 118^\circ$ ii $\tan 242^\circ$

iii $\tan 298^\circ$

c Hence, express each of the following in terms of $\tan x$:

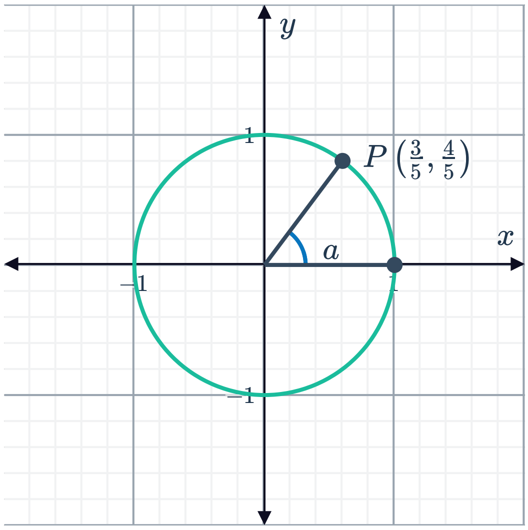
i $\tan\left(180^\circ - x\right)$ ii $\tan\left(180^\circ + x\right)$

iii $\tan\left(360^\circ - x\right)$

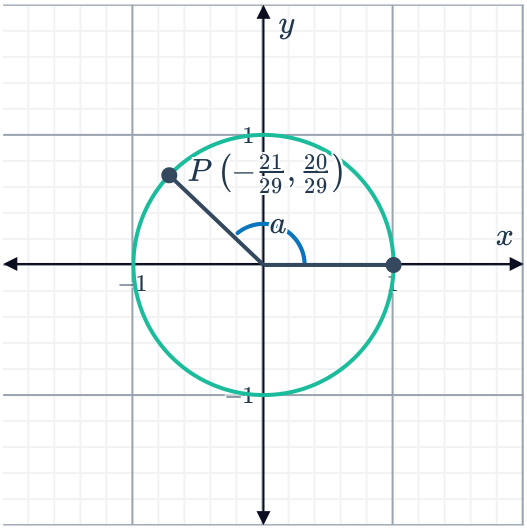
11 For each of the following graphs, find:

i $\sin a$ ii $\cos a$ iii $\tan a$

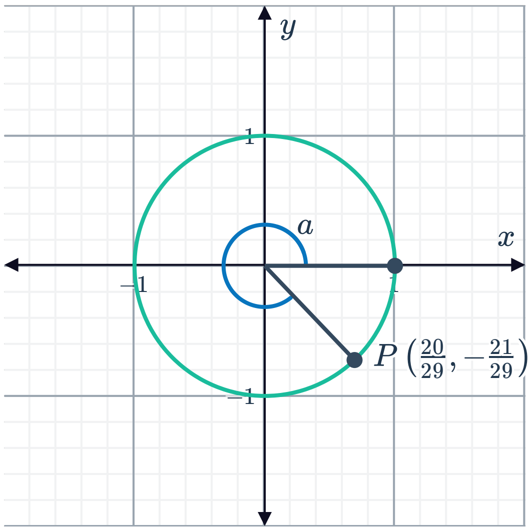
a



b



c



12 The following are angle rotations at point P on the unit circle. Find the acute angle in the first quadrant related to each rotation:

a 295° b -150° c 240° d -75°

13 Rewrite each expression as a trigonometric ratio of a positive acute angle:

a $\sin 93^\circ$ b $\cos 195^\circ$ c $\tan 299^\circ$ d $\sin(-139^\circ)$

e $\cos(-222^\circ)$ f $\tan(-289^\circ)$

14 Simplify:

a $\sin(360^\circ + \theta)$ b $\cos(360^\circ - \theta)$ c $\sin(180^\circ - \theta)$ d $\cos(180^\circ + \theta)$

e $\sin(180^\circ + \theta)$ f $\sin(90^\circ - \theta)$ g $\sin(-\theta)$ h $\cos(-\theta)$

15 Given the approximations $\cos 21^\circ = 0.93$ and $\sin 21^\circ = 0.36$, find the approximate values of the following and giving your answers in two decimal places:

a $\cos 339^\circ$ b $\sin(-339^\circ)$ c $\cos 159^\circ$ d $\sin 159^\circ$

e $\sin 201^\circ$ f $\cos 201^\circ$ g $\sin(-201)^\circ$ h $\cos(-201)^\circ$

16 Evaluate the following correct to two decimal places:

a $\sin 146^\circ$ b $\tan 386^\circ$ c $\cos 387^\circ$

17 The diagram shows P, which represents a rotation of 66° around the unit circle:
Find the following to two decimal places:

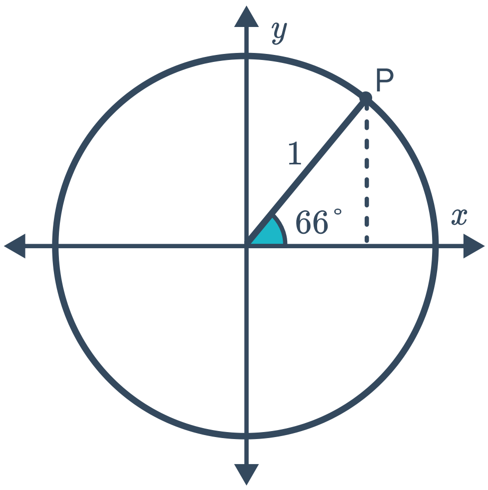
a Coordinates of $P(x, y)$

b The new coordinates of P if it was reflected about the y-axis.

c $\sin 114^\circ$

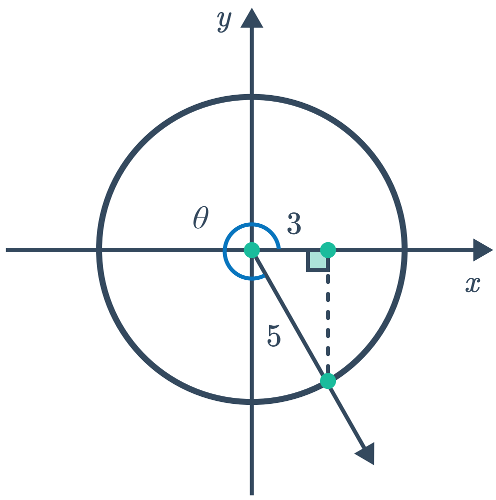
d $\cos 114^\circ$

e $\tan 114^\circ$

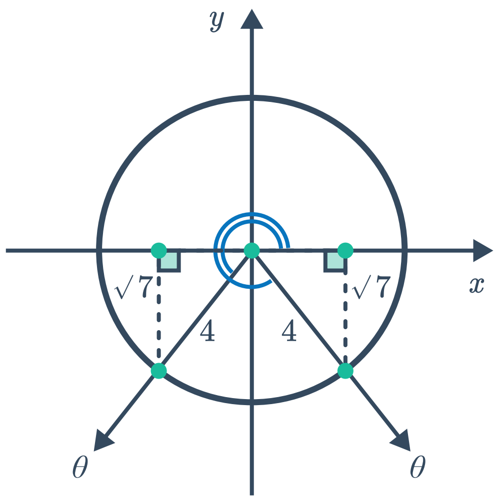


18 Suppose that $\cos\theta = \frac{3}{5}$, where $270^\circ < \theta < 360^\circ$.

a Find $\sin\theta$. b Find $\tan\theta$.



19 Suppose that $\sin\theta = -\dfrac{\sqrt{7}}{4}$.



a Find $\cos\theta$. **b** Find $\tan\theta$.

20 Consider an angle θ such that $\sin\theta = 0.6$ and $\tan\theta < 0$, find:

a $\sin\theta$ **b** $\cos\theta$ **c** $\tan\theta$

21 Consider an angle θ such that $\tan\theta = -1.3$ and $270^\circ < \theta < 360^\circ$, find:

a $\cos\theta$ **b** $\sin\theta$

22 Given that $\tan\theta = -\dfrac{15}{8}$ and $\sin\theta > 0$, find $\cos\theta$.

23 Given the following, find $\sin\theta$.

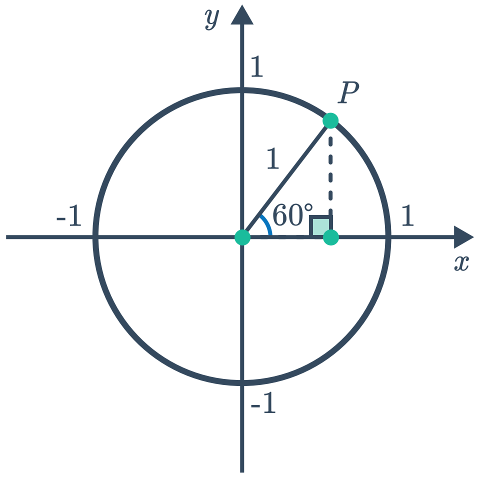
a $\cos\theta = -\dfrac{60}{61}$ and $0^\circ \leq \theta \leq 180^\circ$ **b** $\cos\theta = -\dfrac{6}{7}$ and $\tan\theta < 0$

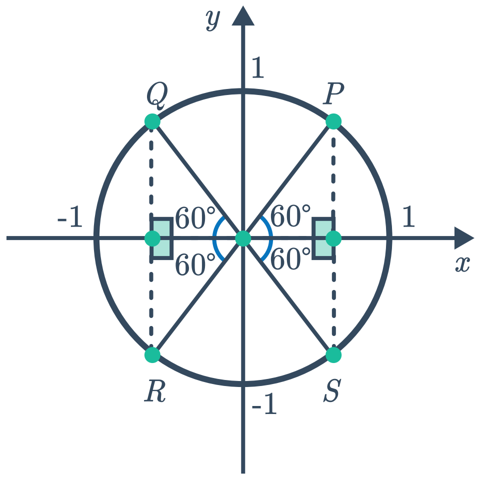
24 Given the following, find the value(s) of $\tan\theta$.

a $\cos\theta = \dfrac{3}{7}$ and θ is acute **b** $\sin\theta = \dfrac{1}{\sqrt{10}}$ and $-90^\circ \leq \theta \leq 90^\circ$

Exact trigonometric ratios

25 The first diagram shows a unit circle with point $P\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$ marked on the circle. Point P represents a rotation of 60° anticlockwise around the origin from the positive x-axis:





a Find the exact values of the following:

i $\sin 60^\circ$ **ii** $\cos 60^\circ$ **iii** $\tan 60^\circ$

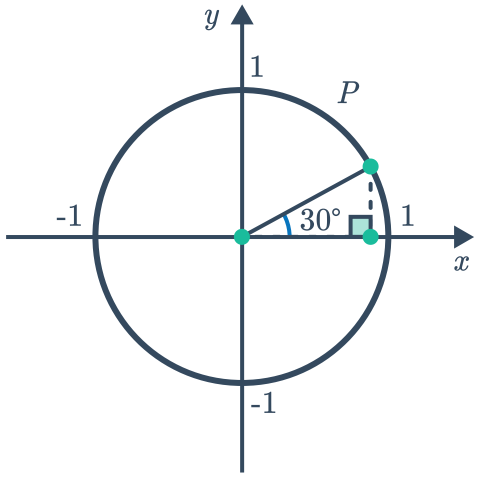
b On the second diagram, the coordinate axes shows a 60° angle that has also been marked in the second, third, and fourth quadrants. For each quadrant, find the relative angle.

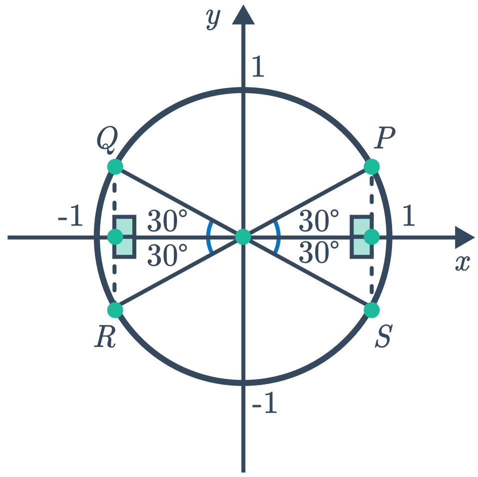
i Quadrant 2 **ii** Quadrant 3 **iii** Quadrant 4

c The points Q, R and S mark rotations of point P to the corresponding angles on the unit circle. Find the exact coordinates of each point:

i Q **ii** R **iii** S

26 The first diagram shows a unit circle with point $P\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$ marked on the circle. Point P represents a rotation of 30° anticlockwise around the origin from the positive x-axis:





a Find the exact values of the following:

i $\sin 30^\circ$ **ii** $\cos 30^\circ$ **iii** $\tan 30^\circ$

b On the second diagram, the coordinate axes shows a 30° angle that has also been marked in the second, third and fourth quadrants. For each quadrant, find the relative angle:

i Quadrant 2 **ii** Quadrant 3 **iii** Quadrant 4

c The points Q, R and S mark rotations of point P to the corresponding angles on the unit circle. Find the exact coordinates of each point:

i Q **ii** R **iii** S

27 Consider the trigonometric ratio $\cos(-210^\circ)$.

a Find the value of the related acute angle.

b Hence, find the value of $\cos(-210^\circ)$ using exact values.

28 Find the exact value of the following:

a $\sin 225^\circ$

b $\cos 225^\circ$

c $\tan 225^\circ$

d $\sin 690^\circ$

e $\sin 135^\circ$

f $\cos 315^\circ$

g $\tan 150^\circ$

h $\sin 240^\circ$

i $\cos 690^\circ$

j $\tan 690^\circ$

k $\sin 50^\circ + \sin 160^\circ + \sin 200^\circ + \sin 310^\circ$

l $\tan 60^\circ + \tan 315^\circ$

29 By rewriting each ratio in terms of the related acute angle, find the exact value of:

$$\frac{\sin 120^\circ \cos 240^\circ \tan 330^\circ}{\tan\left(-45\right)^\circ}$$